Latent Matters: Learning Deep State-Space Models

Presenter: Farzaneh Barat

Alexej Klushyn, Richard Kurle, Botond Cseke, Maximilian Soelch, Patrick van der Smagt

August 24, 2026

Roadmap

1. State-Space Models, Deep State-Space Models (DSSMs), and Variational Autoencoder (VAE)
2. Rate–distortion interpretation of the ELBO.
3. Constrained optimization for better system identification.
4. Variational hierarchical prior (VHP).
5. Extended Kalman VAE (EKVAE).
6. Experiments, model-based control, and conclusions.

From State-Space Models to DSSMs and VAEs

A state-space model represents observations through a latent dynamical state:

$$\mathbf{z}_t \sim p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})$$

$$\mathbf{x}_t \sim p_\theta(\mathbf{x}_t \mid \mathbf{z}_t)$$

For complex and high-dimensional observations, the transition and observation models can be nonlinear:

$$\mathbf{z}_t = f_\psi(\mathbf{z}_{t-1}, \mathbf{u}_{t-1}) + \mathbf{w}_t,$$

$$\mathbf{x}_t = g_\theta(\mathbf{z}_t) + \mathbf{v}_t.$$

where f_ψ and g_θ can be represented by neural networks.

$$\boxed{\text{DSSM} = \text{state-space temporal structure} + \text{nonlinear neural-network models.}}$$

A VAE learns a latent representation $\mathbf{z}_t$ of observations $\mathbf{x}_t$.

$$\boxed{\mathbf{x}_t \xrightarrow[\text{Encoder}]{q_\phi(\mathbf{z}_t|\mathbf{x}_t)} \mathbf{z}_t \xrightarrow[\text{Decoder}]{p_\theta(\mathbf{x}_t|\mathbf{z}_t)} \hat{\mathbf{x}}_t}$$

The posterior $p(\mathbf{z}_t \mid \mathbf{x}_t)$ is generally intractable, the encoder provides approximation: $\Rightarrow q_\phi(\mathbf{z}_t \mid \mathbf{x}_t) \approx p(\mathbf{z}_t \mid \mathbf{x}_t)$
Training maximizes the ELBO:

$$\mathcal{L}_{\text{VAE}} = \overbrace{\mathbb{E}_{q_\phi(\mathbf{z}_t|\mathbf{x}_t)}[\log p(\mathbf{x}_t \mid \mathbf{z}_t)]}^{\text{reconstruction}} - \overbrace{\mathrm{KL}(q_\phi(\mathbf{z}_t \mid \mathbf{x}_t) \,\|\, p(\mathbf{z}_t))}^{\text{latent regularization}}$$

Markov Assumption and DSSM Factorization

Assume that $\mathbf{z}_t$ is a complete Markov state:

$$p(\mathbf{z}_t \mid \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = p(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}).$$

Then the generative model factorizes as

$$p(\mathbf{x}_{1:T}, \mathbf{z}_{1:T} \mid \mathbf{u}_{1:T}) = p(\mathbf{z}_1) \prod_{t=1}^{T} p_\theta(\mathbf{x}_t \mid \mathbf{z}_t) \prod_{t=2}^{T} p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})$$

where $p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})$ describes how the latent state evolves, $p_\theta(\mathbf{x}_t \mid \mathbf{z}_t)$ describes how each latent state generates an observation.

In practice, we observe $\mathbf{x}_{1:T}$ and want to infer the hidden trajectory $\mathbf{z}_{1:T}$:

$$p(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})$$

Using Bayes' rule:

$$p(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) = \frac{p(\mathbf{x}_{1:T}, \mathbf{z}_{1:T} \mid \mathbf{u}_{1:T})}{p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T})}.$$

The denominator requires marginalizing over all possible latent trajectories:

$$p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T}) = \int p(\mathbf{x}_{1:T}, \mathbf{z}_{1:T} \mid \mathbf{u}_{1:T}) \, d\mathbf{z}_{1:T}.$$

For nonlinear neural-network models, the integral is generally intractable.

Sequential ELBO and Rate–Distortion Decomposition

Introduce the approximate posterior:

$$q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) \approx p(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}).$$

$$p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T}) = \int q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) \frac{p(\mathbf{x}_{1:T}, \mathbf{z}_{1:T} \mid \mathbf{u}_{1:T})}{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \, d\mathbf{z}_{1:T}$$

$$p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T}) = \mathbb{E}_{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \left[\frac{p(\mathbf{x}_{1:T}, \mathbf{z}_{1:T} \mid \mathbf{u}_{1:T})}{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \right]$$

Using Jensen's inequality $\Rightarrow \log \mathbb{E}[Y] \geq \mathbb{E}[\log Y]$

$$\log p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T}) \geq \mathcal{F}_{\mathrm{ELBO}} = \mathbb{E}_{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \left[\log \frac{p_\theta(\mathbf{x}_{1:T} \mid \mathbf{z}_{1:T}) p_\psi(\mathbf{z}_{1:T} \mid \mathbf{u}_{1:T})}{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \right]$$

$$\mathcal{F}_{\mathrm{ELBO}} = \mathbb{E}_{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} [\log p_\theta(\mathbf{x}_{1:T} \mid \mathbf{z}_{1:T})] - \mathrm{KL}\Big(q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) \| p_\psi(\mathbf{z}_{1:T} \mid \mathbf{u}_{1:T}) \Big).$$

Define the distortion as follows, measuring the reconstruction error:

$$\mathcal{D}(\theta, \phi) = -\mathbb{E}_{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} [\log p_\theta(\mathbf{x}_{1:T} \mid \mathbf{z}_{1:T})] .$$

Define the rate as follows, measuring the compatibility of the inferred latent trajectory with the transition model:

$$\mathcal{R}(\psi, \phi) = \mathrm{KL}\left(q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) \| p_\psi(\mathbf{z}_{1:T} \mid \mathbf{u}_{1:T}) \right) .$$

$$\boxed{\mathcal{F}_{\mathrm{ELBO}} = -\mathcal{D} - \mathcal{R}.}$$

Why a High ELBO Does Not Guarantee Good Dynamics

The ELBO can be written as

$$\mathcal{F}_{\mathrm{ELBO}} = -\mathcal{D} - \mathcal{R}.$$

Maximizing the ELBO controls the total $\mathcal{D} + \mathcal{R}$, but not the balance between them.
Different rate–distortion pairs can produce the same ELBO:

Model	$\mathcal{D}$	$\mathcal{R}$	$\mathcal{F}_{\mathrm{ELBO}}$
A	100	1	-101
B	50	51	-101

ELBO optimization alone does not guarantee system identification.

Instead of minimizing $\mathcal{D} + \mathcal{R}$ directly, the paper proposes:

$$\boxed{\min_{\theta,\psi,\phi} \mathcal{R}(\psi, \phi) \qquad \text{subject to} \qquad \mathcal{D}(\theta, \phi) \leq \mathcal{D}_0.}$$

$\mathcal{R}$ measures mismatch between the inferred trajectory and the dynamical prior; once reconstruction quality is controlled, reducing $\mathcal{R}$ encourages states consistent with the learned dynamics.

- θ: decoder / observation parameters.
- ψ: transition / dynamics parameters.
- ϕ: recognition / inference parameters.

Solution 1: Constrained Optimization

Introduce a Lagrange multiplier λ to enforce the reconstruction constraint:

$$\boxed{\mathcal{L} = \mathcal{R} + \lambda\big(\mathcal{D} - \mathcal{D}_0\big), \qquad \lambda \geq 0.}$$

Role of λ:

$$\begin{cases} \mathcal{D} > \mathcal{D}_0 & \Rightarrow \lambda \text{ increases and penalizes distortion,} \\ \mathcal{D} \leq \mathcal{D}_0 & \Rightarrow \text{the model can focus more on reducing } \mathcal{R}. \end{cases}$$

Relevant KKT conditions:

$$\mathcal{D} - \mathcal{D}_0 \leq 0, \qquad \lambda \geq 0, \qquad \lambda(\mathcal{D} - \mathcal{D}_0) = 0.$$

$$\boxed{\lambda \text{ automatically controls the balance between } \mathcal{D} \text{ and } \mathcal{R}.}$$

Choice of $\mathcal{D}_0$:

$\mathcal{D}_0$ is the target reconstruction level. It should be achievable but sufficiently strict to ensure good observation fitting. The paper uses the heuristic

$$\boxed{\mathcal{D}_0 = 0.9\,\mathcal{D}_{\text{max}}}$$

The paper first trains the same model using classical variational inference, records its best distortion $\mathcal{D}_{\text{max}}$. The choice of $\mathcal{D}_0$ is a hyperparameter: too strict may make the constraint infeasible, while too loose may allow poor reconstruction.

Lower-Bound Condition: $0 \leq \lambda \leq 1$

The constrained objective is

$$\mathcal{L} = \mathcal{R} + \lambda(\mathcal{D} - \mathcal{D}_0)$$

For $\lambda > 0$, minimizing $\mathcal{L}$ is equivalent to

$$\min \left(\mathcal{D} + \frac{1}{\lambda}\mathcal{R} \right)$$

Define $\beta = \frac{1}{\lambda}$, so the equivalent maximization objective is

$$\mathcal{F}_\beta = -\mathcal{D} - \beta\mathcal{R}$$

The standard ELBO is

$$\mathcal{F}_{\mathrm{ELBO}} = -\mathcal{D} - \mathcal{R} \leq \log p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T}).$$

Therefore,

$$\mathcal{F}_\beta = \mathcal{F}_{\mathrm{ELBO}} - (\beta - 1)\mathcal{R}$$

Since

$$\mathcal{R} = \mathrm{KL}\left(q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}, \mathbf{u}) \| p_\psi(\mathbf{z}_{1:T} \mid \mathbf{u})\right) \geq 0$$

The lower-bound property is guaranteed if

$$\beta \geq 1 \quad \Longleftrightarrow \quad \frac{1}{\lambda} \geq 1 \quad \Longleftrightarrow \quad \boxed{0 < \lambda \leq 1}$$

Including $\lambda = 0$ as the limiting boundary. The lower-bound guarantee is required at the end of training.

$$\boxed{0 \leq \lambda \leq 1}$$

Problem 2: Mismatch of the Initial-State Prior

Many DSSMs use

$$p(\mathbf{z}_1) = \mathcal{N}(\mathbf{0}, \mathbf{I})$$

However, the initial latent states inferred from the data may have a non-Gaussian distribution.

Define the aggregated initial-state posterior as the distribution of initial latent states over the whole dataset.

$$q_{\text{agg}}(\mathbf{z}_1) = \mathbb{E}_{p_{\mathcal{D}}(\mathbf{x},\mathbf{u})}\left[q_\phi(\mathbf{z}_1 \mid \mathbf{x}, \mathbf{u})\right].$$

We want to choose a prior $p(\mathbf{z}_1)$ to minimize its average mismatch with the inferred initial-state posteriors across the dataset:

$$J[p] = \mathbb{E}_{p_{\mathcal{D}}}\left[\mathrm{KL}\left(q_\phi(\mathbf{z}_1 \mid \mathbf{x}, \mathbf{u}) \,\|\, p(\mathbf{z}_1)\right)\right].$$

$$J[p] = \mathbb{E}_{p_{\mathcal{D}}}\left[\int q_\phi(\mathbf{z}_1 \mid \mathbf{x}, \mathbf{u}) \log q_\phi(\mathbf{z}_1 \mid \mathbf{x}, \mathbf{u})\, d\mathbf{z}_1\right] - \mathbb{E}_{p_{\mathcal{D}}}\left[\int q_\phi(\mathbf{z}_1 \mid \mathbf{x}, \mathbf{u}) \log p(\mathbf{z}_1)\, d\mathbf{z}_1\right].$$

The first term is independent of $p(\mathbf{z}_1)$, so define it as a constant C_1:

$$J[p] = C_1 - \int \mathbb{E}_{p_{\mathcal{D}}(\mathbf{x},\mathbf{u})}\left[q_\phi(\mathbf{z}_1 \mid \mathbf{x}, \mathbf{u})\right] \log p(\mathbf{z}_1)\, d\mathbf{z}_1 = C + \mathrm{KL}(q_{\text{agg}}(\mathbf{z}_1) \,\|\, p(\mathbf{z}_1)).$$

Since $\mathrm{KL}(q \| p) \geq 0$, the minimum occurs when

$$p^*(\mathbf{z}_1) = q_{\text{agg}}(\mathbf{z}_1)$$

Therefore, the best prior is the distribution of initial latent states actually inferred from the dataset.

Solution 2: Variational Hierarchical Prior (VHP)

The optimal prior may be highly non-Gaussian, so the paper introduces the hierarchical prior

$$\boxed{p_{\psi_0}(\mathbf{z}_1) = \int p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)\, d\zeta, \qquad p(\zeta) = \mathcal{N}(\mathbf{0}, \mathbf{I}).}$$

Although ζ is Gaussian, $p_{\psi_0}(\mathbf{z}_1 \mid \zeta)$ parameters are produced by a nonlinear neural network, so the marginal $p_{\psi_0}(\mathbf{z}_1)$ can be highly non-Gaussian. Thus, VHP increases the flexibility of the initial-state distribution.

The integral is difficult to optimize directly. Therefore, introduce an approximate posterior

$$q_{\phi_0}(\zeta \mid \mathbf{z}_1).$$

Then, by Jensen's inequality,

$$\log p_{\psi_0}(\mathbf{z}_1) = \log \mathbb{E}_{q_{\phi_0}(\zeta|\mathbf{z}_1)} \left[\frac{p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)}{q_{\phi_0}(\zeta \mid \mathbf{z}_1)} \right] \geq \underbrace{\mathbb{E}_{q_{\phi_0}(\zeta|\mathbf{z}_1)} \left[\log \frac{p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)}{q_{\phi_0}(\zeta \mid \mathbf{z}_1)} \right]}_{\mathcal{F}_{\mathrm{VHP}}(\psi_0, \phi_0; \mathbf{z}_1)}.$$

An expectation under $p^*(\mathbf{z}_1)$ is equivalent to averaging over the dataset and the corresponding posterior q_ϕ

$$\begin{aligned}
\mathbb{E}_{p^*(\mathbf{z}_1)}[\log p_{\psi_0}(\mathbf{z}_1)] &= \mathbb{E}_{p_{\mathcal{D}}(\mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \mathbb{E}_{q_\phi(\mathbf{z}_1 | \mathbf{x}_{1:T}, \mathbf{u}_{1:T})}[\log p_{\psi_0}(\mathbf{z}_1)] \\
&= \mathbb{E}_{p_{\mathcal{D}}(\mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \mathbb{E}_{q_\phi(\mathbf{z}_{1:T} | \mathbf{x}_{1:T}, \mathbf{u}_{1:T})}[\log p_{\psi_0}(\mathbf{z}_1)] \\
&\geq \mathbb{E}_{p_{\mathcal{D}}(\mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \mathbb{E}_{q_\phi(\mathbf{z}_{1:T} | \mathbf{x}_{1:T}, \mathbf{u}_{1:T})}[\mathcal{F}_{\mathrm{VHP}}(\psi_0, \phi_0; \mathbf{z}_1)]
\end{aligned}$$

VHP: Lower Bound on the Prior and Upper Bound on the Rate

Because $\mathcal{F}_{\mathrm{VHP}}(\psi_0, \phi_0; \mathbf{z}_1) \leq \log p_{\psi_0}(\mathbf{z}_1)$ replacing $\log p_{\psi_0}(\mathbf{z}_1)$ introduces an upper bound on the rate:

$$
\begin{aligned}
\mathcal{R}(\psi, \phi, \psi_0) &= \mathbb{E}_{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \left[\log q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) - \log p_{\psi_0}(\mathbf{z}_1) - \sum_{t=2}^{T} \log p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) \right] \\
&\leq \mathbb{E}_{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})} \left[\log q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) - \mathcal{F}_{\mathrm{VHP}}(\psi_0, \phi_0; \mathbf{z}_1) - \sum_{t=2}^{T} \log p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) \right] \\
&= \mathcal{R}(\psi, \phi, \psi_0, \phi_0).
\end{aligned}
$$

Therefore,

$$\mathcal{R}(\psi, \phi, \psi_0) \leq \mathcal{R}(\psi, \phi, \psi_0, \phi_0),$$

so the VHP rate is conservative and still gives a lower bound on the ELBO.
The previous bound leads to the Lagrangian

$$\mathcal{L}(\theta, \psi, \phi, \psi_0, \phi_0; \lambda) = \mathcal{R}(\psi, \phi, \psi_0, \phi_0) + \lambda \left(\mathcal{D}(\theta, \phi) - \mathcal{D}_0 \right).$$

The corresponding constrained optimization problem is

$$\overbrace{\min_{\psi_0, \phi_0}}^{\substack{\text{Empirical} \\ \text{Bayes}}} \; \overbrace{\min_{\theta, \psi}}^{\text{M-step}} \; \overbrace{\max_{\lambda} \; \min_{\phi}}^{\text{E-step}} \; \mathcal{L}(\theta, \psi, \phi, \psi_0, \phi_0; \lambda) \qquad \text{s.t.} \qquad \lambda \geq 0.$$

REWO Update for the Lagrange Multiplier

The constrained objective is

$$\mathcal{L} = \mathcal{R}(\psi, \phi, \psi_0, \phi_0) + \lambda\big(\mathcal{D}(\theta, \phi) - \mathcal{D}_0\big).$$

Because the batch distortion is noisy, Algorithm 1 does not use the instantaneous value directly. First, it computes an exponentially smoothed distortion:

$$\widehat{\mathcal{D}}^{(i)} = (1-\alpha)\widehat{\mathcal{D}}^{(i)}_{\mathrm{ba}} + \alpha\widehat{\mathcal{D}}^{(i-1)} \quad \text{with} \quad \widehat{\mathcal{D}}^{(0)} = \widehat{\mathcal{D}}^{(0)}_{\mathrm{ba}}.$$

Define the reconstruction-constraint error

$$\delta^{(i)} = \widehat{\mathcal{D}}^{(i)} - \mathcal{D}_0.$$

The Lagrange multiplier is updated multiplicatively:

$$\lambda^{(i)} = \lambda^{(i-1)} \exp\left[-\nu\, f_\lambda\left(\lambda^{(i-1)}, \delta^{(i)}; \tau_1, \tau_2\right) \delta^{(i)}\right]$$

The feedback function is

$$f_\lambda(\lambda, \delta; \tau_1, \tau_2) = \big(1 - H(\delta)\big) \tanh\left[\tau_1 \left(\frac{1}{\lambda} - 1\right)\right] - \tau_2 H(\delta), \qquad H(\delta) = \begin{cases} 1, & \delta > 0, \\ 0, & \delta \le 0. \end{cases}$$

Important

The lower-bound guarantee requires $0 \le \lambda \le 1$ at the end of training; λ may temporarily exceed 1 while the reconstruction constraint is violated. Eq. (8) is a feedback rule, not a hard projection: λ may exceed 1 during early training, but after the reconstruction constraint is satisfied it is driven back toward 1.

Two Training Phases in REWO

Phase 1: Initial phase

While
$$\widehat{\mathcal{D}}^{(i)} > \mathcal{D}_0,$$
the reconstruction constraint is violated.
Only the encoder and decoder parameters are optimized:
$$(\theta, \phi) \quad \text{are updated.}$$
The transition model and VHP are not yet trained:
$$\boxed{\psi,\ \psi_0,\ \phi_0 \quad \text{are fixed.}}$$
Since $\delta^{(i)} > 0$, we have
$$H(\delta) = 1, \quad f_\lambda = -\tau_2, \quad \Rightarrow \quad \lambda^{(i)} = \lambda^{(i-1)} e^{\nu \tau_2 \delta^{(i)}}$$
Thus,
$$\lambda^{(i)} > \lambda^{(i-1)}$$
and reconstruction receives increasing weight.

Phase 2: Main phase

As soon as
$$\widehat{\mathcal{D}}^{(i)} \leq \mathcal{D}_0,$$
the reconstruction quality is acceptable.
Training now includes all model parameters:
$$\boxed{(\theta, \psi, \phi, \psi_0, \phi_0) \quad \text{are updated.}}$$
The model can therefore start learning

transition dynamics (ψ) and initial-state prior (ψ_0, ϕ_0)

After Phase 1, typically $\lambda > 1$. For $\delta \leq 0$ and $\lambda > 1$,
$$f_\lambda = \tanh\left[\tau_1 \left(\frac{1}{\lambda} - 1\right)\right] < 0,$$
which drives λ downward.

$$\boxed{\text{reconstruction first} \longrightarrow \text{dynamics second.}}$$

Algorithm

Algorithm 1 REWO for deep state-space models

Initialise $i = 1$
Initialise $\lambda^{(0)} = 1$
Initialise $\textsc{InitialPhase} = \textsc{True}$
while training **do**
  Compute $\hat{\mathcal{D}}_{\text{ba}}$ (batch average)
  $\hat{\mathcal{D}}^{(i)} = (1-\alpha) \cdot \hat{\mathcal{D}}_{\text{ba}} + \alpha \cdot \hat{\mathcal{D}}^{(i-1)}, \quad (\hat{\mathcal{D}}^{(0)} = \hat{\mathcal{D}}_{\text{ba}})$
  $\lambda^{(i)} \leftarrow \lambda^{(i-1)} \cdot \exp\left[-\nu \cdot f_\lambda\left(\lambda^{(i-1)}, \hat{\mathcal{D}}^{(i)} - \mathcal{D}_0; \tau_1, \tau_2\right) \cdot \left(\hat{\mathcal{D}}^{(i)} - \mathcal{D}_0\right)\right]$
  if $\hat{\mathcal{D}}^{(i)} \leq \mathcal{D}_0$ **then**
    $\textsc{InitialPhase} = \textsc{False}$
  end if
  if $\textsc{InitialPhase}$ **then**
    Optimise $\mathcal{L}(\theta, \psi, \phi, \psi_0, \phi_0; \lambda^{(i)})$ w.r.t. θ, ϕ
  else
    Optimise $\mathcal{L}(\theta, \psi, \phi, \psi_0, \phi_0; \lambda^{(i)})$ w.r.t. $\theta, \psi, \phi, \psi_0, \phi_0$
  end if
  $i \leftarrow i + 1$
end while

Motivation: Extended Kalman VAE (EKVAE)

Limitation for dynamical systems:

A standard VAE treats observations independently and does not explicitly model latent dynamics.
For time-series data, temporal dependencies between latent states should be modeled.

$$\mathbf{z}_{t-1} \longrightarrow \mathbf{z}_t \longrightarrow \mathbf{z}_{t+1}$$

Consider a pendulum with two different types of information.

$$\text{Position information: } \phi_t, \quad \text{Dynamic information: } \dot{\phi}_t$$

- A single frame may provide enough information to reconstruct the current image, but not enough to determine the dynamics, such as $\dot{\phi}_t$.
- A VAE learns how to encode and reconstruct each observation, while a DSSM additionally learns how the latent state evolves over time.
- Most DSSMs use RNNs to approximate filtering/smoothing or inside the transition model; this can limit dynamics accuracy and can leave part of the state information in the RNN rather than in $\mathbf{z}_t$.

EKVAE idea

Introduce an auxiliary encoded observation $\mathbf{a}_t$:

$$\boxed{\mathbf{x}_t \xrightarrow{\text{Encoder NN}} \mathbf{a}_t \xrightarrow{\text{Kalman smoothing}} \mathbf{z}_t \xrightarrow{\text{dynamics}} \mathbf{z}_{t+1}}$$

$$\mathbf{x}_t \xrightarrow{q_\phi(\mathbf{a}_t|\mathbf{x}_t)} \mathbf{a}_t \xrightarrow{p_\psi(\mathbf{z}_t|\mathbf{a}_{1:T},\mathbf{u}_{1:T-1})} \mathbf{z}_t \xrightarrow{p_\psi(\mathbf{z}_{t+1}|\mathbf{z}_t,\mathbf{u}_t)} \mathbf{z}_{t+1}$$

EKVAE Architecture and Inference

EKVAE uses a hierarchical latent representation:

$$\zeta \to \mathbf{z}_1 \to \mathbf{z}_2 \to \cdots \to \mathbf{z}_T,$$

with

$$\mathbf{z}_t \to \mathbf{a}_t \to \mathbf{x}_t.$$

Directly mapping $\mathbf{z}_t$ to high-dimensional images is too nonlinear for efficient Kalman inference. Therefore, EKVAE separates the model into:

- VAE encoder/decoder between $\mathbf{a}_t$ and $\mathbf{x}_t$;

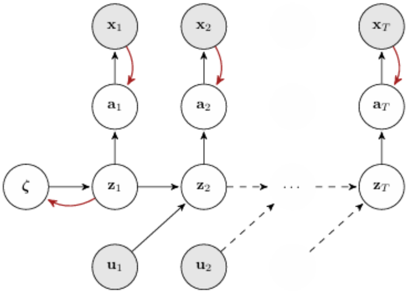
$$\mathbf{x}_t \xrightarrow{\text{Encoder NN}} q_\phi(\mathbf{a}_t \mid \mathbf{x}_t), \quad \mathbf{a}_t \xrightarrow{\text{Decoder NN}} p_\theta(\mathbf{x}_t \mid \mathbf{a}_t)$$

- linear-Gaussian measurement model between $\mathbf{z}_t$ and $\mathbf{a}_t$;

$$p_\psi(\mathbf{a}_t \mid \mathbf{z}_t) = \mathcal{N}(\mathbf{a}_t \mid \mathbf{H}\mathbf{z}_t, \mathbf{R})$$

- Kalman filtering/smoothing in the low-dimensional latent space.

The Gaussian structure enables efficient Kalman inference.



- $\mathbf{x}_t \leftrightarrow \mathbf{a}_t$: VAE encoder/decoder maps between observations and auxiliary latent variables.
- $\mathbf{z}_t \to \mathbf{a}_t$: linear-Gaussian measurement model enables Kalman inference.
- $\mathbf{z}_t \to \mathbf{z}_{t+1}$: locally linear transition model describes the latent dynamics.
- $\zeta \to \mathbf{z}_1$: VHP provides a flexible prior for the initial latent state.

Locally Linear Transition Model

The nonlinear dynamical system is modeled by a Gaussian transition model that is locally linear w.r.t. discrete time steps:

$$\boxed{p_\psi(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{u}_t) = \mathcal{N}\left(\mathbf{z}_{t+1} \mid \mathbf{F}_\psi(\mathbf{z}_t, \mathbf{u}_t)\mathbf{z}_t + \mathbf{B}_\psi(\mathbf{z}_t, \mathbf{u}_t)\mathbf{u}_t, \mathbf{Q}_\psi(\mathbf{z}_t, \mathbf{u}_t)\right).}$$

Each matrix is a convex mixture of M weighted base matrices:

$$\mathbf{F}_\psi(\mathbf{z}_t, \mathbf{u}_t) = \sum_{m=1}^{M} \alpha_\psi^{(m)}(\mathbf{z}_t, \mathbf{u}_t)\mathbf{F}^{(m)}.$$

Similarly,

$$\mathbf{B}_\psi(\mathbf{z}_t, \mathbf{u}_t) = \sum_{m=1}^{M} \alpha_\psi^{(m)}(\mathbf{z}_t, \mathbf{u}_t)\mathbf{B}^{(m)}, \qquad \mathbf{Q}_\psi(\mathbf{z}_t, \mathbf{u}_t) = \sum_{m=1}^{M} \alpha_\psi^{(m)}(\mathbf{z}_t, \mathbf{u}_t)\mathbf{Q}^{(m)}.$$

The base matrices $\mathbf{F}^{(m)}, \mathbf{B}^{(m)}, \mathbf{Q}^{(m)}$ are learnable transition parameters and are optimized jointly with the neural network through the overall training objective. The mixture weights are produced by a neural network:

$$\boxed{\boldsymbol{\alpha}_\psi(\mathbf{z}_t, \mathbf{u}_t) = \mathrm{softmax}\left(\mathbf{g}_\psi(\mathbf{z}_t, \mathbf{u}_t)\right).}$$

Therefore,

$$\alpha_\psi^m \geq 0, \qquad \sum_{m=1}^{M} \alpha_\psi^m = 1.$$

Hence, softmax makes $\mathbf{F}_\psi$, $\mathbf{B}_\psi$, and $\mathbf{Q}_\psi$ convex interpolations of several learned local models.
Softmax provides normalized mixture weights, but does not by itself guarantee dynamical stability.

EKF Prediction: Local Taylor Linearization

Consider the nonlinear dynamics

$$\mathbf{z}_t = \mathbf{f}(\mathbf{z}_{t-1}, \mathbf{u}_{t-1}) + \mathbf{q}_{t-1}, \qquad \mathbf{q}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_{t-1}).$$

Given the filtered state

$$p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-1}) = \mathcal{N}(\mathbf{m}_{t-1}, \mathbf{P}_{t-1}),$$

the EKF predicts

$$p_\psi(\mathbf{z}_t \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-1}) = \mathcal{N}(\mathbf{m}_t^-, \mathbf{P}_t^-).$$

The predicted mean is

$$\boxed{\mathbf{m}_t^- = \mathbf{f}(\mathbf{m}_{t-1}, \mathbf{u}_{t-1})}$$

For covariance propagation, the EKF locally linearizes $\mathbf{f}$ using a first-order Taylor expansion:

$$\boxed{\mathbf{F}_{t-1} = \left.\frac{\partial \mathbf{f}(\mathbf{z}, \mathbf{u})}{\partial \mathbf{z}}\right|_{\mathbf{m}_{t-1}, \mathbf{u}_{t-1}}}$$

Thus,

$$\boxed{\mathbf{P}_t^- = \mathbf{F}_{t-1}\mathbf{P}_{t-1}\mathbf{F}_{t-1}^\top + \mathbf{Q}_{t-1}.}$$

Key idea: standard EKF requires evaluating the Jacobian of the nonlinear dynamics at every time step.

EKVAE: Neural Linearization

Instead of explicitly computing the Jacobians of an unknown nonlinear function $\mathbf{f}$, EKVAE directly learns state- and input-dependent local linear operators.
The transition model is

$$\boxed{p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) = \mathcal{N}\left(\mathbf{z}_t \mid \mathbf{F}_\psi \mathbf{z}_{t-1} + \mathbf{B}_\psi \mathbf{u}_{t-1}, \mathbf{Q}_\psi\right),}$$

where

$$\mathbf{F}_\psi = \mathbf{F}_\psi(\mathbf{z}_{t-1}, \mathbf{u}_{t-1}), \qquad \mathbf{B}_\psi = \mathbf{B}_\psi(\mathbf{z}_{t-1}, \mathbf{u}_{t-1}).$$

They play the role of the EKF Jacobians:

$$\boxed{\frac{\partial \mathbf{f}}{\partial \mathbf{z}} \longrightarrow \mathbf{F}_\psi(\mathbf{z}, \mathbf{u}), \qquad \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \longrightarrow \mathbf{B}_\psi(\mathbf{z}, \mathbf{u}).}$$

Evaluated at the filtered mean,

$$\mathbf{m}_t^- = \mathbf{F}_\psi(\mathbf{m}_{t-1}, \mathbf{u}_{t-1})\mathbf{m}_{t-1} + \mathbf{B}_\psi(\mathbf{m}_{t-1}, \mathbf{u}_{t-1})\mathbf{u}_{t-1},$$
$$\mathbf{F}_{t-1} = \mathbf{F}_\psi(\mathbf{m}_{t-1}, \mathbf{u}_{t-1}), \qquad \mathbf{Q}_{t-1} = \mathbf{Q}_\psi(\mathbf{m}_{t-1}, \mathbf{u}_{t-1}),$$
$$\mathbf{P}_t^- = \mathbf{F}_{t-1}\mathbf{P}_{t-1}\mathbf{F}_{t-1}^\top + \mathbf{Q}_{t-1}.$$

Neural linearization

The model learns the local linearization from data instead of computing a Taylor-expansion Jacobian at every time step.

Kalman Filtering and RTS Smoothing

Kalman Filter: Forward Update

Given the prediction

$$p(\mathbf{z}_t \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-1}) = \mathcal{N}(\mathbf{m}_t^-, \mathbf{P}_t^-),$$

The auxiliary measurement model is linear:

$$\mathbf{a}_t = \mathbf{H}\mathbf{z}_t + \mathbf{r}_t, \qquad \mathbf{r}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

Compute the innovation, covariance, and Kalman gain

$$\mathbf{e}_t = \mathbf{a}_t - \mathbf{H}\mathbf{m}_t^-,$$

$$\mathbf{S}_t = \mathbf{H}\mathbf{P}_t^-\mathbf{H}^\top + \mathbf{R}.$$

$$\mathbf{K}_t = \mathbf{P}_t^-\mathbf{H}^\top\mathbf{S}_t^{-1}$$

Then update the latent state:

$$\mathbf{m}_t = \mathbf{m}_t^- + \mathbf{K}_t\mathbf{e}_t$$

$$\mathbf{P}_t = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_t^-$$

RTS Smoother: Backward Update

Filtering only uses observations up to time t:

$$p(\mathbf{z}_t \mid \mathbf{a}_{1:t}, \mathbf{u}_{1:t-1}).$$

Smoothing uses the **entire sequence**:

$$p(\mathbf{z}_t \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1}).$$

Starting from the filtered estimates, define

$$\mathbf{G}_t = \mathbf{P}_t\mathbf{F}_t^\top(\mathbf{P}_{t+1}^-)^{-1},$$

where $\mathbf{G}_t$ is the RTS smoother gain.
The backward corrections are

$$\mathbf{m}_t^s = \mathbf{m}_t + \mathbf{G}_t(\mathbf{m}_{t+1}^s - \mathbf{m}_{t+1}^-)$$

$$\mathbf{P}_t^s = \mathbf{P}_t + \mathbf{G}_t(\mathbf{P}_{t+1}^s - \mathbf{P}_{t+1}^-)\mathbf{G}_t^\top$$

Disentangling Static and Dynamic Features

The matrix $\mathbf{H}$ is chosen as a rectangular identity matrix:

$$\mathbf{H} = (\delta_{ij}) \in \mathbb{R}^{D_a \times D_z}.$$

Example:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

The first D_a state dimensions correspond to features encoded by $\mathbf{a}_t$; the remaining dimensions capture information inferred from temporal evolution.

The key idea is to separate latent information into two parts:

1. **Static features** can be inferred from a single observation.
2. **The auxiliary variable is linked to the first dimensions of the state:**

$$\mathbf{a}_t \approx \mathbf{H}\mathbf{z}_t, \quad \mathbf{H} = \begin{bmatrix} \mathbf{I}_{D_a} & \mathbf{0} \end{bmatrix}.$$

3. **The remaining dimensions are inferred from temporal information:**

$$\mathbf{a}_{1:T} \xrightarrow{\text{dynamics + Kalman smoother}} \text{dynamic components of } \mathbf{z}_t$$

Thus,

$$\underbrace{\text{single-frame information}}_{\text{e.g., position}} \to \text{first latent dimensions}, \qquad \underbrace{\text{temporal information}}_{\text{e.g., velocity}} \to \text{remaining latent dimensions.}$$

EKVAE Generative Model: Appendix A.6.1

Any DSSM whose ELBO can be written as $\mathcal{F}_{\text{ELBO}} = -\mathcal{D} - \mathcal{R}$ can use CO. For each model:

derive ELBO $\rightarrow$ identify distortion $\mathcal{D}$ $\rightarrow$ identify rate $\mathcal{R}$ $\rightarrow$ apply constrained optimization.

To derive the EKVAE ELBO, we start from the marginal likelihood and integrate out all latent variables.

$$\begin{aligned}
p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T}) &= \iiint p(\mathbf{x}_{1:T}, \mathbf{a}_{1:T}, \mathbf{z}_{1:T}, \zeta \mid u_{1:T})\, d\mathbf{a}_{1:T}\, d\mathbf{z}_{1:T}\, d\zeta \\
&= \iiint p(\mathbf{x}_{1:T} \mid \mathbf{a}_{1:T}) p_\psi(\mathbf{a}_{1:T} \mid \mathbf{z}_{1:T}) p_\psi(\mathbf{z}_{1:T} \mid \zeta, \mathbf{u}_{1:T}) p(\zeta)\, d\mathbf{a}_{1:T}\, d\mathbf{z}_{1:T}\, d\zeta \\
&= \iiint \prod_{t=1}^{T} [p_\theta(\mathbf{x}_t \mid \mathbf{a}_t) p_\psi(\mathbf{a}_t \mid \mathbf{z}_t)] \prod_{t=2}^{T} p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)\, d\mathbf{a}_{1:T}\, d\mathbf{z}_{1:T}\, d\zeta.
\end{aligned}$$

Introduce the auxiliary encoder:

$$q_\phi(\mathbf{a}_{1:T} \mid \mathbf{x}_{1:T}).$$

Then Jensen's inequality gives:

$$\log p(\mathbf{x}_{1:T} \mid \mathbf{u}_{1:T}) \geq \mathbb{E}_{q_\phi(\mathbf{a}_{1:T} \mid x_{1:T})} \left[\log \frac{p_\theta(\mathbf{x}_{1:T} \mid a_{1:T})}{q_\phi(\mathbf{a}_{1:T} \mid \mathbf{x}_{1:T})} + \log p(\mathbf{a}_{1:T} \mid \mathbf{u}_{1:T}) \right].$$

$$p_\theta(\mathbf{x}_{1:T} \mid \mathbf{a}_{1:T}) = \prod_{t=1}^{T} p_\theta(\mathbf{x}_t \mid \mathbf{a}_t), \quad \Rightarrow \quad -\mathcal{D}(\theta, \phi) = \sum_{t=1}^{T} \mathbb{E}_{q_\phi(\mathbf{a}_t \mid \mathbf{x}_t)} [\log p_\theta(\mathbf{x}_t \mid \mathbf{a}_t)].$$

Chain Factorization of the Auxiliary Likelihood

$$
\begin{aligned}
p(\mathbf{a}_{1:T} \mid \mathbf{u}_{1:T}) &= \iint \prod_{t=1}^{T} p_\psi(\mathbf{a}_t \mid \mathbf{z}_t) \prod_{t=2}^{T} p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) p_{\psi_0}(\mathbf{z}_1 \mid \zeta)\, p(\zeta)\, d\mathbf{z}_{1:T}\, d\zeta \\
&= \iint \prod_{t=2}^{T} \left[p_\psi(\mathbf{a}_t \mid \mathbf{z}_t) p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) \right] p_\psi(\mathbf{a}_1 \mid \mathbf{z}_1) p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)\, d\mathbf{z}_{1:T}\, d\zeta.
\end{aligned}
$$

The integral over $\mathbf{z}_{1:T}$ using the Markov structure rewrites as an initial term plus sequential filtering terms:

$$
\begin{aligned}
\log p(\mathbf{a}_{1:T} \mid \mathbf{u}_{1:T}) = \log \iint & p_\psi(\mathbf{a}_1 \mid \mathbf{z}_1) p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)\, d\mathbf{z}_1\, d\zeta \\
&+ \sum_{t=2}^{T} \log \iint \underbrace{p_\psi(\mathbf{a}_t \mid \mathbf{z}_t)}_{\text{observation}} \underbrace{p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})}_{\text{transition}} \underbrace{p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-2})}_{\text{filtered distribution}}\, d\mathbf{z}_t\, d\mathbf{z}_{t-1}
\end{aligned}
$$

The integral over $\mathbf{z}_t$ is solved analytically because the EKVAE uses locally linear Gaussian dynamics.

$$
\int p_\psi(\mathbf{a}_t \mid \mathbf{z}_t) p_\psi(z_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})\, dz_t = p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}).
$$

Monte Carlo Marginalization Using the Smoother

After integrating out $\mathbf{z}_t$ analytically, we obtain $p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})$, but we must still marginalize over $\mathbf{z}_{t-1}$:

$$p_\psi(\mathbf{a}_t \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-1}) = \int p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})\, p_f(\mathbf{z}_{t-1})\, d\mathbf{z}_{t-1},$$

The filtered and smoothed distribution are:

$$p_f(\mathbf{z}_{t-1}) = p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-2}), \qquad p_s(\mathbf{z}_{t-1}) = p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1}).$$

Multiply and divide by $p_s(\mathbf{z}_{t-1})$:

$$p_\psi(\mathbf{a}_t \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-1}) = \int p_s(\mathbf{z}_{t-1}) \frac{p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) p_f(\mathbf{z}_{t-1})}{p_s(\mathbf{z}_{t-1})}\, d\mathbf{z}_{t-1} = \mathbb{E}_{p_s(\mathbf{z}_{t-1})}\left[\frac{p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) p_f(\mathbf{z}_{t-1})}{p_s(\mathbf{z}_{t-1})}\right].$$

Now take the logarithm and apply Jensen's inequality:

$$\log p_\psi(\mathbf{a}_t \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-1}) \geq \mathbb{E}_{p_s(\mathbf{z}_{t-1})}\left[\log p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) + \log \frac{p_f(\mathbf{z}_{t-1})}{p_s(\mathbf{z}_{t-1})}\right].$$

Why use the smoother?

Samples from p_s use the complete sequence and provide better state estimates for sample-based optimization of the transition model.

Initial-State Term: Introducing the VHP

For the initial time step, we start from

$$\log \iint p(\mathbf{a}_1 \mid \mathbf{z}_1)\, p(\mathbf{z}_1 \mid \zeta)\, p(\zeta)\, d\mathbf{z}_1\, d\zeta.$$

Introduce the smoothed state distribution: $p_s(z_1) = p(\mathbf{z}_1 \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1})$
and the variational distribution $q(\zeta \mid \mathbf{z}_1)$.
Multiplying and dividing by $p_s(\mathbf{z}_1)q(\zeta \mid \mathbf{z}_1)$ gives

$$\log \iint p_s(\mathbf{z}_1)q(\zeta \mid \mathbf{z}_1)\frac{p(\mathbf{a}_1 \mid z_1)p(\mathbf{z}_1 \mid \zeta)p(\zeta)}{p_s(\mathbf{z}_1)q(\zeta \mid \mathbf{z}_1)}\, d\mathbf{z}_1\, d\zeta.$$

Using Jensen's inequality,

$$\geq \mathbb{E}_{p_s(\mathbf{z}_1)}\left[\log p(\mathbf{a}_1 \mid \mathbf{z}_1) - \log p_s(\mathbf{z}_1) + \mathbb{E}_{q(\zeta|\mathbf{z}_1)}\left[\log \frac{p(\mathbf{z}_1 \mid \zeta)p(\zeta)}{q_\phi(\zeta \mid \mathbf{z}_1)}\right]\right].$$

The last expectation is exactly the VHP lower bound:

$$\mathcal{F}_{\mathrm{VHP}}(\psi_0, \phi_0; \mathbf{z}_1) = \mathbb{E}_{q_{\phi_0}(\zeta|\mathbf{z}_1)}\left[\log \frac{p_{\psi_0}(\mathbf{z}_1 \mid \zeta)p(\zeta)}{q_{\phi_0}(\zeta \mid \mathbf{z}_1)}\right].$$

Complete EKVAE Smoother ELBO: Appendix A.6.2

$$
\begin{aligned}
\log p(x_{1:T} \mid u_{1:T}) \geq \mathbb{E}_{q_\phi(a_{1:T}\mid x_{1:T})} \Bigg[& \log \frac{p_\theta(x_{1:T} \mid a_{1:T})}{q_\phi(a_{1:T} \mid x_{1:T})} \\
& + \mathbb{E}_{p_\psi(z_1 \mid a_{1:T}, u_{1:T-1})} \Big[\log p_\psi(a_1 \mid z_1) - \log p_\psi(z_1 \mid a_{1:T}, u_{1:T-1}) + \mathcal{F}_{\mathrm{VHP}} \Big] \\
& + \sum_{t=2}^{T} \mathbb{E}_{p_\psi(z_{t-1} \mid a_{1:T}, u_{1:T-1})} \Bigg[\log p_\psi(a_t \mid z_{t-1}, u_{t-1}) + \log \frac{\overbrace{p_\psi(z_{t-1} \mid a_{1:t-1}, u_{1:t-2})}^{\text{filtered distribution}}}{\underbrace{p_\psi(z_{t-1} \mid a_{1:T}, u_{1:T-1})}_{\text{smoothed distribution}}} \Bigg] \Bigg] \\
= & \underbrace{\sum_{t=1}^{T} \mathbb{E}_{q_\phi(\mathbf{a}_t \mid \mathbf{x}_t)} \left[\log p_\theta(\mathbf{x}_t \mid \mathbf{a}_t) \right]}_{-D(\boldsymbol{\theta}, \phi)} - \mathbb{E}_{q_\phi(\mathbf{a}_{1:T} \mid \mathbf{u}_{1:T})} \left[\mathbb{E}_{p_\psi(\mathbf{z}_1 \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1})} \mathcal{R}_{\mathrm{initial}} \right] \\
& + \sum_{t=2}^{T} \mathbb{E}_{p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1})} \left[\log \frac{q_\phi(\mathbf{a}_t \mid \mathbf{x}_t)}{p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})} + \log \frac{p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1})}{p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-2})} \right].
\end{aligned}
$$

Initial-State Rate in the Smoother ELBO

The paper defines the initial-state rate as

$$\mathcal{R}_{\text{initial}}(\psi, \phi, \psi_0, \phi_0; \mathbf{a}_{1:T}, \mathbf{z}_1) = \log \frac{q_\phi(\mathbf{a}_1 \mid \mathbf{x}_1)}{p_\psi(\mathbf{a}_1 \mid \mathbf{z}_1)} + \log p_\psi(\mathbf{z}_1 \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1}) - \mathcal{F}_{\text{VHP}}(\psi_0, \phi_0; \mathbf{z}_1).$$

The VHP lower bound is

$$\mathcal{F}_{\text{VHP}}(\psi_0, \phi_0; \mathbf{z}_1) = \mathbb{E}_{q_{\phi_0}(\zeta \mid \mathbf{z}_1)}\left[\log \frac{p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)}{q_{\phi_0}(\zeta \mid \mathbf{z}_1)}\right].$$

Substituting $\mathcal{F}_{\text{VHP}}$ gives

$$\mathcal{R}_{\text{initial}}(\psi, \phi, \psi_0, \phi_0; \mathbf{a}_{1:T}, \mathbf{z}_1) = \log \frac{q_\phi(\mathbf{a}_1 \mid \mathbf{x}_1)}{p_\psi(\mathbf{a}_1 \mid \mathbf{z}_1)} + \mathbb{E}_{q_{\phi_0}(\zeta \mid \mathbf{z}_1)}\left[\log \frac{p_\psi(\mathbf{z}_1 \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1})}{p_{\psi_0}(\mathbf{z}_1 \mid \zeta)} + \log \frac{q_{\phi_0}(\zeta \mid \mathbf{z}_1)}{p(\zeta)}\right].$$

$$\mathcal{R}(\psi, \phi, \psi_0, \phi_0) = \mathbb{E}_{q(\mathbf{z}_{1:T}, \mathbf{a}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T})}\Bigg[\mathcal{R}_{\text{initial}}\left(\psi, \phi, \psi_0, \phi_0; \mathbf{a}_{1:T}, \mathbf{z}_1\right) + \sum_{t=2}^{T}\left(\log \frac{q_\phi(\mathbf{a}_t \mid \mathbf{x}_t)}{p_\psi(\mathbf{a}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})} + \log \frac{p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:T}, \mathbf{u}_{1:T-1})}{p_\psi(\mathbf{z}_{t-1} \mid \mathbf{a}_{1:t-1}, \mathbf{u}_{1:t-2})}\right)\Bigg].$$

Initial-State Rate in the Smoother ELBO

$$\mathbb{E}_{q_\phi(\mathbf{a}_1|\mathbf{x}_1)}\left[\log\frac{q_\phi(\mathbf{a}_1 \mid \mathbf{x}_1)}{p_\psi(\mathbf{a}_1 \mid \mathbf{z}_1)}\right] = \mathrm{KL}(q_\phi(\mathbf{a}_1 \mid \mathbf{x}_1) \| p_\psi(\mathbf{a}_1 \mid \mathbf{z}_1))$$

$$\mathbb{E}_{p_s(\mathbf{z}_1)}\left[\log\frac{p_s(\mathbf{z}_1)}{p_{\psi_0}(\mathbf{z}_1 \mid \zeta)}\right] = \mathrm{KL}(p_s(\mathbf{z}_1) \| p_{\psi_0}(\mathbf{z}_1 \mid \zeta))$$

$$\mathbb{E}_{q_{\phi_0}(\zeta|\mathbf{z}_1)}\left[\log\frac{q_{\phi_0}(\zeta \mid \mathbf{z}_1)}{p(\zeta)}\right] = \mathrm{KL}(q_{\phi_0}(\zeta \mid \mathbf{z}_1) \| p(\zeta))$$

$$\begin{aligned}
&\mathcal{F}^{\mathrm{EKVAE,smooth}}_{\mathrm{ELBO}} \\
&\approx \mathbb{E}_{z_{1:T}, a_{1:T} \sim q(z_{1:T}, a_{1:T} | x_{1:T}, u_{1:T})} \mathbb{E}_{\zeta \sim q(\zeta | z_1)} \Bigg[\sum_{t=1}^{T} \log p_\theta(x_t \mid a_t) \\
&- \Big(\mathrm{KL}(q_\phi(a_1 \mid x_1) \| p_\psi(a_1 \mid z_1)) + \mathrm{KL}(p_\psi(z_1 \mid a_{1:T}, u_{1:T-1}) \| p_{\psi_0}(z_1 \mid \zeta)) + \mathrm{KL}(q_{\phi_0}(\zeta \mid z_1) \| p(\zeta)) \Big) \\
&- \sum_{t=2}^{T} \Big(\mathrm{KL}(q_\phi(a_t \mid x_t) \| p_\psi(a_t \mid z_{t-1}, u_{t-1})) + \mathrm{KL}(p_\psi(z_{t-1} \mid a_{1:T}, u_{1:T-1}) \| p_\psi(z_{t-1} \mid a_{1:t-1}, u_{1:t-2})) \Big) \Bigg]
\end{aligned}$$

EKVAE Filter ELBO: Appendix A.6.3

The filter version replaces the smoothed distribution with the filtered distribution:

$$p_\psi(z_t \mid a_{1:T}, u_{1:T-1}) \quad \longrightarrow \quad p_\psi(z_t \mid a_{1:t}, u_{1:t-1}).$$

Key consequence: The filter version enables sample-based optimization of the transition parameters ψ.

$$\log p(x_{1:T} \mid u_{1:T}) \geq \sum_{t=1}^{T} \mathbb{E}_{q_\phi(a_t|x_t)} [\log p_\theta(x_t \mid a_t)] - \mathbb{E}_{q_\phi(a_{1:T}|x_{1:T})} \Bigg[\mathbb{E}_{p_\psi(z_1|a_1)} \mathbb{E}_{q_{\phi_0}(\zeta|z_1)} \left[\log \frac{q_\phi(a_1 \mid x_1)}{p_\psi(a_1 \mid z_1)} + \log \frac{p_\psi(z_1 \mid a_1)}{p_{\psi_0}(z_1 \mid \zeta)} + \log \frac{q_{\phi_0}(\zeta \mid z_1)}{p(\zeta)} \right] + \sum_{t=2}^{T} \mathbb{E}_{p_\psi(z_{t-1}|a_{1:t-1}, u_{1:t-2})} \left[\log \frac{q_\phi(a_t \mid x_t)}{p_\psi(a_t \mid z_{t-1}, u_{t-1})} \right] \Bigg].$$

$$\mathcal{F}_{\mathrm{ELBO}}^{\mathrm{EKVAE,filter}}$$

$$\approx \mathbb{E}_{z_{1:T}, a_{1:T} \sim q} \mathbb{E}_{\zeta \sim q_{\phi_0}(\zeta|z_1)} \left[\sum_{t=1}^{T} \log p_\theta(x_t \mid a_t) - \Big(\mathrm{KL}(q_\phi(a_1 \mid x_1) \| p_\psi(a_1 \mid z_1)) + \mathrm{KL}(p_\psi(z_1 \mid a_1) \| p_{\psi_0}(z_1 \mid \zeta)) + \mathrm{KL}(q_{\phi_0}(\zeta \mid z_1) \| p(\zeta)) \Big) - \sum_{t=2}^{T} \mathrm{KL} \left(q_\phi(a_t \mid x_t) \| p_\psi(a_t \mid z_{t-1}, u_{t-1}) \right) \right].$$

Interpretation: The smoother uses future observations to infer z_t, while the filter only uses information available up to time t.

Applying CO to Existing DSSM Baselines

The proposed constrained-optimization framework is **not specific to EKVAE**.
For any DSSM whose ELBO can be written as

$$\mathcal{F}_{\mathrm{ELBO}} = -\mathcal{D} - \mathcal{R},$$

$$\mathcal{L} = \mathcal{R} + \lambda(\mathcal{D} - \mathcal{D}_0).$$

DKS / VHP-DKS

- It uses an **RNN recognition model** to approximate the smoothed posterior from the whole sequence:

$$\mathbf{x}_{1:T} \xrightarrow{\text{LSTM}} q_\phi(\mathbf{z}_t \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}).$$

- The latent dynamics are modeled separately by

$$p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}),$$

  so the rate measures the mismatch between the **RNN-inferred state** and the **transition-model prediction**.

- **VHP-DKS keeps the same inference, decoder, and dynamics;** only the initial-state prior is replaced:

$$\mathcal{N}(\mathbf{0}, \mathbf{I}) \quad \longrightarrow \quad p_{\psi_0}(\mathbf{z}_1).$$

DVBS / VHP-DVBS

- **DVBS = Deep Variational Bayes Smoother.**
- It also uses a **neural/RNN-based recognition model**, but its transition dynamics are **locally linear**:

$$p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1}) = \mathcal{N}(\mathbf{F}_t \mathbf{z}_{t-1} + \mathbf{B}_t \mathbf{u}_{t-1}, \mathbf{Q}_t).$$

- The original DVBS constructs the initial state through an inferred pseudo-state:

$$\zeta \sim q_{\phi_0}(\zeta \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}), \qquad \mathbf{z}_1 = f_{\psi_0}(\zeta).$$

- **VHP-DVBS replaces this deterministic construction** with a flexible probabilistic prior:

$$p_{\psi_0}(\mathbf{z}_1) = \int p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta)\, d\zeta.$$

Datasets and Evaluation Metrics

Pendulum

- 500 sequences.
- 15 images per sequence.
- Image size: 16×16.
- True state includes angle and angular velocity.

Reacher

- 2000 sequences.
- 30 time steps per sequence.
- Two observation settings: angles or RGB images.
- RGB image size: $64 \times 64 \times 3$.

The paper evaluates models using two criteria:
1. System identification Measured by linear regression R^2.
For pendulum:

$$R^2_{\mathrm{angle}} = \frac{R^2_{\sin\phi} + R^2_{\cos\phi}}{2}, \qquad R^2_{\mathrm{velocity}} = R^2_{\dot{\phi}}$$

2. Prediction accuracy

$\boxed{\text{Good dynamics} \Rightarrow \text{high } R^2 \text{ and low prediction MSE.}}$

CO Training Dynamics on DKS

Apply the proposed training method to an existing model to show that the proposed CO framework can improve a DSSM without changing its basic architecture.

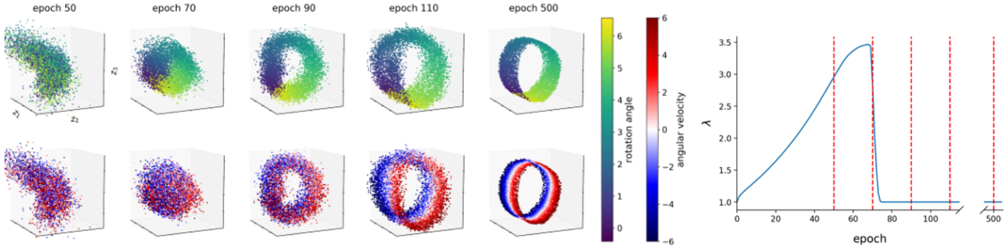


Figure 1: VHP-DKS (CO) on pendulum (image data). Learning the state-space representation with the CO framework. Distortion and rate are balanced by the Lagrange multiplier λ, which is updated (cf. Alg. 1) such that the model first improves the reconstruction quality/constraint by learning the rotation angle (see epoch 70). As soon as the constraint is satisfied, λ decreases and the model starts learning the underlying dynamics, i.e. to represent the angular velocity. Robustness w.r.t. the hyperparameter $\mathcal{D}_0$ is demonstrated in App. A.7.1 (Fig. 9).

Similar ELBO, Very Different Dynamics

On pendulum images, different models have similar ELBO values, but very different dynamics.

Table 1: The CO framework significantly improves system identification, as indicated by the high correlation (R^2 of an OLS regression) between inferred and ground-truth states. This leads to an increased prediction accuracy of the model, measured by the MSE of 500 predicted sequences conditioned on $\{\mathbf{z}_1^{(n)} \sim q(\mathbf{z}_1 \,|\, \mathbf{x}_{1:5}^{(n)}, \mathbf{u}_{1:4}^{(n)})\}_{n=1}^{500}$. Note that high ELBO values do not imply the model can accurately predict the observed system.

(a) Pendulum (image data)

model	test ELBO	$R^{2\ (\text{ang. } \phi)}_{\text{OLS reg.}}$	$R^{2\ (\text{vel. } \dot{\phi})}_{\text{OLS reg.}}$	$\text{MSE}^{\ (\text{smoothed})}_{\text{predict}}$
VHP-EKVAE (CO)	**807.3**	**0.992**	**0.998**	**1.99E-4**
EKVAE (CO)	805.9	0.957	0.991	3.53E-4
EKVAE (annealing)	804.2	0.687	0.339	1.94E-3
VHP-DVBS (CO)	804.3	**0.992**	0.989	5.63E-4
DVBS (CO)	803.8	0.985	0.980	9.41E-4
DVBS (annealing)	803.1	0.795	0.237	4.67E-3
VHP-DKS (CO)	804.7	0.973	0.990	1.73E-3
DKS (CO)	804.1	0.912	0.962	2.36E-3
DKS (annealing)	804.0	0.330	0.040	2.12E-2

Bold indicates the best result

Red indicates a low correlation with the ground truth

(b) Reacher (angle data)

model	$R^{2\ (\text{ang. } \phi)}_{\text{OLS reg.}}$	$R^{2\ (\text{ang. } \psi)}_{\text{OLS reg.}}$	$R^{2\ (\text{vel. } \dot{\phi})}_{\text{OLS reg.}}$	$R^{2\ (\text{vel. } \dot{\psi})}_{\text{OLS reg.}}$	$\text{MSE}^{\ (\text{smoothed})}_{\text{predict}}$
VHP-EKVAE (CO)	0.988	**0.997**	**0.989**	0.986	**2.13E-5**
EKVAE (annealing)	0.712	0.835	0.881	0.339	4.38E-4
VHP-DVBS (CO)	**0.990**	0.994	0.979	**0.991**	2.75E-4
DVBS (annealing)	0.897	0.949	0.963	0.778	4.17E-4
VHP-DKS (CO)	0.984	0.991	0.986	0.980	3.52E-4
DKS (annealing)	0.693	0.781	0.965	0.016	1.12E-3

(c) Reacher (RGB image data)

model	$R^{2\ (\text{ang. } \phi)}_{\text{OLS reg.}}$	$R^{2\ (\text{ang. } \psi)}_{\text{OLS reg.}}$	$R^{2\ (\text{vel. } \dot{\phi})}_{\text{OLS reg.}}$	$R^{2\ (\text{vel. } \dot{\psi})}_{\text{OLS reg.}}$	$\text{MSE}^{\ (\text{smoothed})}_{\text{predict}}$
VHP-EKVAE (CO)	**0.980**	**0.986**	**0.991**	**0.987**	**1.64E-4**
EKVAE (annealing)	0.672	0.052	0.668	0.091	1.82E-3

Importantly, CO is also compared with standard training using the same baseline architecture; therefore the improvement cannot be attributed only to the increased capacity of EKVAE.

VHP Fixes the Initial-Prior Mismatch

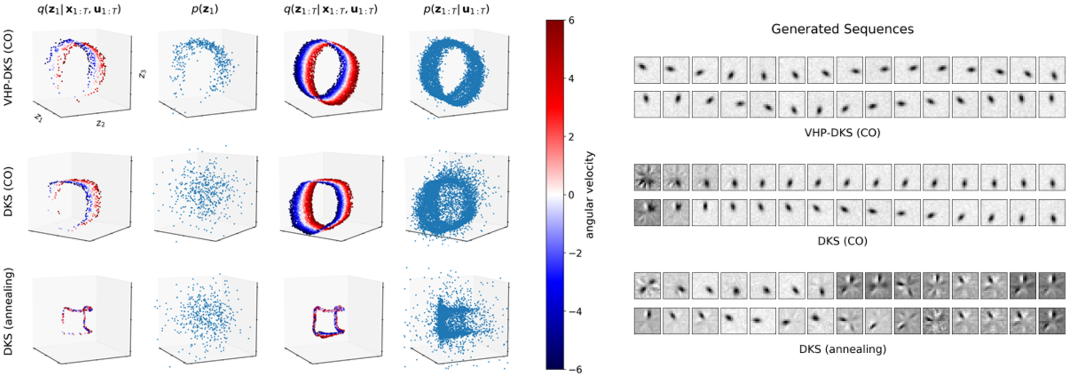


Figure 2: Pendulum (image data). In contrast to annealing (bottom), CO (middle & top) enables the model to learn the underlying dynamic system, as we verify in Table 1a. Furthermore, the VHP (top) significantly improves the quality of generated sequences. This is because the VHP learns a prior $p(\mathbf{z}_1) = \mathbb{E}_{p(\boldsymbol{\zeta})}\left[p(\mathbf{z}_1 | \boldsymbol{\zeta})\right]$ that matches the manifold of $\mathbb{E}_{p_{\mathcal{D}}}\left[q(\mathbf{z}_1 | \mathbf{x}_{1:T}, \mathbf{u}_{1:T})\right]$ (cf. first two columns of the latent-space visualisations).

Using the Learned Latent State for Control

The EKVAE not only learns a latent state for prediction. It can also organize the state into physically meaningful parts.

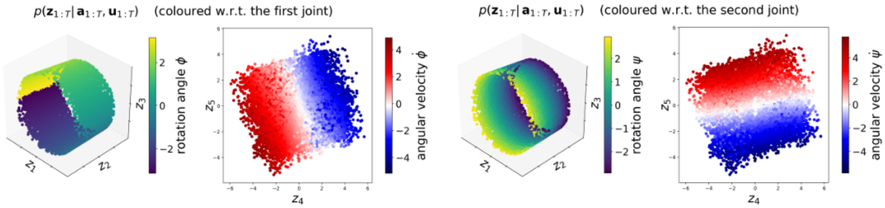


Figure 3: VHP-EKVAE (CO) on reacher (RGB image data). Five-dimensional state-space representation (disentangled): the first three dimensions (z_1, z_2, z_3) represent the two joint angles, the last two dimensions (z_4, z_5) represent the respective angular velocities. The barrel shape indicates the model has learned that the first joint can do a 360 degree turn; whereas the second joint is restricted to avoid self-collisions (cf. Fig. 7).

Limitation of RNN-Based Transition Models

In RNN-based DSSMs, part of the dynamical information may be stored in the deterministic RNN hidden state $\mathbf{h}_t$ rather than in $\mathbf{z}_t$.
For the KVAE,

$$p(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{h}_t, \mathbf{u}_t), \qquad \mathbf{h}_t = \mathrm{LSTM}(\mathbf{a}_{1:t}).$$

For the RSSM,

$$p(\mathbf{z}_{t+1} \mid \mathbf{h}_t), \qquad \mathbf{h}_t = \mathrm{LSTM}(\mathbf{z}_{1:t}, \mathbf{u}_{1:t}).$$

Therefore, the future may depend on

$$(\mathbf{z}_t, \mathbf{h}_t)$$

instead of $\mathbf{z}_t$ alone.

$\boxed{\mathbf{z}_t \text{ may no longer be a complete Markov state.}}$

For the pendulum, the experiments show that $\mathbf{z}_t$ strongly represents the angle but has very low correlation with the true angular velocity.

How does the paper verify this?
Compare predictions initialized from:

$$\textbf{Smoothed initial state:} \qquad \mathbf{z}_1 \sim p(\mathbf{z}_1 \mid \mathbf{a}_{1:5}, \mathbf{u}_{1:4}),$$

with

$$\mathbf{h}_1 = \mathrm{LSTM}(\mathbf{a}_1),$$

versus

$$\textbf{Filtered state at } t = 5: \qquad \mathbf{z}_5 \sim p(\mathbf{z}_5 \mid \mathbf{a}_{1:5}, \mathbf{u}_{1:4}),$$

with

$$\mathbf{h}_5 = \mathrm{LSTM}(\mathbf{a}_{1:5}).$$

At $t = 1$, the smoother gives $\mathbf{z}_1$ access to the full sequence, but $\mathbf{h}_1$ has seen only one observation.
If angular velocity were stored in $\mathbf{z}_1$, smoothing-based predictions should be accurate. Instead, filtering-based predictions are much better.

$$\Rightarrow \quad \text{velocity information is mainly stored in } \mathbf{h}_t.$$

Reacher Control from an Encoded Goal Position

Why EKVAE helps

EKVAE removes the deterministic RNN path and performs Bayesian filtering/smoothing directly in $\mathbf{z}_t$, encouraging the latent state itself to contain the information required to describe the dynamics.

Table 2: Pendulum (image data), cf. Tab. 1a. As a consequence of the RNN-based transition models, the angular velocity is not encoded in $\mathbf{z}_t$ but in the RNN. This is indicated by the low correlation (R^2) with the ground-truth and verified by the low accuracy of the smoothing-based predictions, as we explain in Fig. 4 on the example of the KVAE.

model	$R^{2\ (\text{ang. } \phi)}_{\text{OLS reg.}}$	$R^{2\ (\text{vel. } \dot{\phi})}_{\text{OLS reg.}}$	$\text{MSE}^{\text{(smoothed)}}_{\text{predict}}$	$\text{MSE}^{\text{(filtered)}}_{\text{predict}}$
VHP-KVAE (CO)	0.989	0.043	2.87E-3	4.24E-4
KVAE (annealing)	0.652	0.134	3.16E-3	6.67E-4
VHP-RSSM (CO)	0.915	0.086	3.10E-3	4.80E-4
RSSM (annealing)	0.158	0.060	3.23E-3	6.94E-4

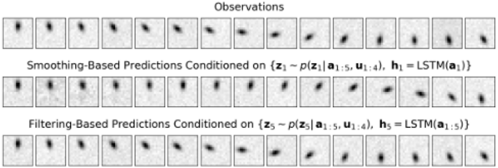
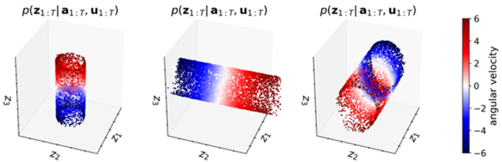


Figure 4: VHP-KVAE (CO). The predictions show that the KVAE encodes the angular velocity of the pendulum in $\mathbf{h}_t = \text{LSTM}(\mathbf{a}_{1:t})$ and not in $\mathbf{z}_t$. This causes the poor smoothing-based predictions, as $\mathbf{h}_1 = \text{LSTM}(\mathbf{a}_1)$ does not have access to sequence data and therefore cannot infer the angular velocity.

Reacher Control from an Encoded Goal Position



$\mathbf{H} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ $\mathbf{H} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\mathbf{H} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Figure 5: VHP-EKVAE (CO) on pendulum (image data). Disentangled (position–velocity) state-space representations: $\mathbf{H}$ defines the latent dimensions where the model learns to encode the rotation angle and the angular velocity.

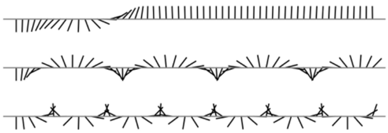


Figure 6: VHP-EKVAE (CO). Visualisation of different policies that we learned based on the disentangled position–velocity representation in Fig. 5 (left): pendulum swing-up (top) and steady rotation (middle & bottom) by encoding the goal position for $r_t^{\text{pos}}(\mathbf{z}_t, \mathbf{p}_g)$ and the goal angular velocity for $r_t^{\text{vel}}(\mathbf{z}_t, \mathbf{v}_g)$, respectively.

Figure 7: VHP-EKVAE (CO). Policy for reaching an encoded goal position (red dot) that we learned based on the disentangled (position–velocity) state-space representation in Fig. 3. This verifies that the EKVAE learns an accurate model of the reacher including self-collision avoidance.

Conclusions

The experiments support three main claims:

1. CO improves system identification in the reported experiments, including comparisons where the underlying DSSM architecture is fixed.
2. VHP improves the initial latent prior.
3. EKVAE learns better Markov states.

Conclusions:

- A high ELBO does not necessarily guarantee good latent dynamics.
- The paper shows that reconstruction and dynamics must be controlled separately through the rate–distortion.
- Constrained optimization first enforces acceptable reconstruction, then improves the dynamics by minimizing the rate.
- VHP improves the initial latent prior by matching it to the aggregated posterior instead of using a fixed Gaussian prior.
- EKVAE combines auxiliary VAEs with Kalman filtering/smoothing to obtain a structured posterior for DSSMs.

Thank You!

Questions?

Baseline 1: DKS and VHP-DKS under CO

DKS uses an RNN to **approximate smoothing**. For the DKS, the ELBO is written as

$$\boxed{\mathcal{F}_{\mathrm{ELBO}}^{\mathrm{DKS}}(\theta,\psi,\phi) = -\mathcal{D}_{\mathrm{DKS}}(\theta,\phi) - \mathcal{R}_{\mathrm{DKS}}(\psi,\phi)}$$

The distortion is the negative expected reconstruction likelihood:

$$\mathcal{D}_{\mathrm{DKS}} = -\sum_{t=1}^{T} \mathbb{E}_{q_\phi(\mathbf{z}_t \mid \mathbf{x}_{1:T},\mathbf{u}_{1:T})}\left[\log p_\theta(\mathbf{x}_t \mid \mathbf{z}_t)\right].$$

The rate regularizes the inferred latent trajectory against the dynamics prior:

$$\mathcal{R}_{\mathrm{DKS}} = \mathrm{KL}(q_\phi(\mathbf{z}_1 \mid \mathbf{x}_{1:T},\mathbf{u}_{1:T}) \| p(\mathbf{z}_1)) + \sum_{t=2}^{T} \mathbb{E}_{q_\phi(\mathbf{z}_{t-1} \mid \mathbf{x}_{1:T},\mathbf{u}_{1:T})}\Big[\mathrm{KL}(q_\phi(\mathbf{z}_t \mid \mathbf{z}_{t-1},\mathbf{x}_{t:T},\mathbf{u}_{t-1:T}) \| p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1},\mathbf{u}_{t-1}))\Big].$$

VHP-DKS keeps the same reconstruction term: $\mathcal{D}_{\mathrm{VHP-DKS}} = \mathcal{D}_{\mathrm{DKS}}$.
Only the initial prior term changes:

$$\mathrm{KL}(q_\phi(\mathbf{z}_1 \mid \mathbf{x},\mathbf{u}) \| p(\mathbf{z}_1)) \quad \longrightarrow \quad \mathbb{E}_{q_\phi(\mathbf{z}_1)}\left[\log q_\phi(\mathbf{z}_1 \mid \mathbf{x},\mathbf{u}) - \mathcal{F}_{\mathrm{VHP}}(\psi_0,\phi_0;\mathbf{z}_1)\right].$$

$$\begin{aligned}\mathcal{R}_{\mathrm{VHP-DKS}} =& \mathbb{E}_{q_\phi(\mathbf{z}_1 \mid \mathbf{x}_{1:T},\mathbf{u}_{1:T})}\left[\log q_\phi(\mathbf{z}_1 \mid \mathbf{x}_{1:T},\mathbf{u}_{1:T}) - \mathcal{F}_{\mathrm{VHP}}(\psi_0,\phi_0;\mathbf{z}_1)\right] \\ &+ \sum_{t=2}^{T} \mathbb{E}_{q_\phi(\mathbf{z}_{t-1} \mid \mathbf{x}_{1:T},\mathbf{u}_{1:T})}\Big[\mathrm{KL}(q_\phi(\mathbf{z}_t \mid \mathbf{z}_{t-1},\mathbf{x}_{t:T},\mathbf{u}_{t-1:T}) \| p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1},\mathbf{u}_{t-1}))\Big].\end{aligned}$$

Baseline 2: DVBS/DVBF under CO

DVBS uses an empirical-Bayes-style initial state:

$$\zeta \sim q_{\phi_0}(\zeta \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}), \qquad \mathbf{z}_1 = f_{\psi_0}(\zeta).$$

Approximate posterior factorization:

$$q_\phi(\mathbf{z}_{2:T} \mid f_{\psi_0}(\zeta), \mathbf{x}_{2:T}, \mathbf{u}_{1:T}) = q_\phi(\mathbf{z}_2 \mid f_{\psi_0}(\zeta), \mathbf{x}_{2:T}, \mathbf{u}_{1:T}) \prod_{t=3}^{T} q_\phi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{x}_{t:T}, \mathbf{u}_{t-1:T}).$$

Distortion:

$$\mathcal{D}_{\mathrm{DVBS}} = -\mathbb{E}_{q_{\phi_0}} \mathbb{E}_{q_\phi} \left[\log p_\theta(\mathbf{x}_1 \mid f_{\psi_0}(\zeta)) + \sum_{t=2}^{T} \log p_\theta(\mathbf{x}_t \mid \mathbf{z}_t) \right].$$

Rate:

$$\begin{aligned}\mathcal{R}_{\mathrm{DVBS}} = \mathbb{E}_{q_{\phi_0}} \mathbb{E}_{q_\phi} \Big[&\mathrm{KL}(q_{\phi_0}(\zeta \mid \mathbf{x}, \mathbf{u}) \| p(\zeta)) \\ &+ \mathrm{KL}(q_\phi(\mathbf{z}_2 \mid f_{\psi_0}(\zeta), \mathbf{x}_{2:T}, \mathbf{u}_{1:T}) \| p_\psi(\mathbf{z}_2 \mid f_{\psi_0}(\zeta), \mathbf{u}_1)) \\ &+ \sum_{t=3}^{T} \mathrm{KL}(q_\phi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{x}_{t:T}, \mathbf{u}_{t-1:T}) \| p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})) \Big].\end{aligned}$$

$$\boxed{\mathcal{L}_{\mathrm{DVBS}} = \mathcal{R}_{\mathrm{DVBS}} + \lambda(\mathcal{D}_{\mathrm{DVBS}} - \mathcal{D}_0).}$$

Baseline 2 with VHP: VHP-DVBS

With VHP, the deterministic initial state

$$\mathbf{z}_1 = f_{\psi_0}(\zeta)$$

is replaced by a probabilistic initial prior:

$$p_{\psi_0}(\mathbf{z}_1) = \int p_{\psi_0}(\mathbf{z}_1 \mid \zeta) p(\zeta) d\zeta.$$

The posterior becomes uniform over all time steps:

$$q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) = q_\phi(\mathbf{z}_1 \mid \mathbf{x}_{1:T}, \mathbf{u}_{1:T}) \prod_{t=2}^{T} q_\phi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{x}_{t:T}, \mathbf{u}_{t-1:T}).$$

Distortion:

$$\mathcal{D}_{\mathrm{VHP-DVBS}} = -\mathbb{E}_{q_\phi(\mathbf{z}_{1:T} \mid \mathbf{x}, \mathbf{u})} \left[\sum_{t=1}^{T} \log p_\theta(\mathbf{x}_t \mid \mathbf{z}_t) \right].$$

Rate:

$$\begin{aligned} \mathcal{R}_{\mathrm{VHP-DVBS}} =& \mathbb{E}_{q_\phi(\mathbf{z}_1)} [\log q_\phi(\mathbf{z}_1 \mid \mathbf{x}, \mathbf{u}) - \mathcal{F}_{\mathrm{VHP}}(\psi_0, \phi_0; \mathbf{z}_1)] \\ & + \sum_{t=2}^{T} \mathbb{E}_{q_\phi(\mathbf{z}_{t-1})} \left[\mathrm{KL}(q_\phi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{x}_{t:T}, \mathbf{u}_{t-1:T}) \| p_\psi(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{u}_{t-1})) \right]. \end{aligned}$$